Bambusa reports positive phase 1 results for BBT001 (anti-IL-4Rα/IL-31 bispecific) in AtD


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Summary

Bambusa reported preliminary proof-of-concept (POC) data from the ongoing global phase 1 trial of BBT001, a half-life-extended bisp antibody targeting IL-4Rα and IL-31, in bio-naive patients with moderate-to-severe atopic dermatitis (AtD). In a small cohort (n=17; 12 placebo), BBT001 450 mg IV Q2W for 4 weeks produced highly statistically significant placebo-adjusted EASI reductions beginning at (–35.56%, p=0.0012), deepening through week 6 (–78.95%, p<0.0001). Rapid itch relief was observed from day 1, no conjunctivitis wa and extended half-life supports potential Q3M maintenance dosing. Phase 2b planned; additional IV data (12-week AtD bio-naïve/bio- 14-week CSU) expected 1H'27

Why it Matters: BBT001 is the only IL-4Rα/IL-31 bispecific antibody in active clinical development after J&J terminated its IL-4Rα/IL-3 did not meet the "high bar" of efficacy in a phase 2b study for atopic dermatitis. The positive efficacy is encouraging but not surprisin IL-4R and IL-31 are validated target. However, it is too early to claim BBT001 is the "best in disease" based on data from 17 patients. A control in the future phase 2b study like the one of J&J will be imperative to show the true potential of BBT001

Key Highlights

- BBT001 is a novel tetravalent "2+2" bispecific antibody engineered to simultaneously target the IL-4/IL-13 and IL-31 pathways, fe modifications to extend its half-life
- Efficacy results: Data cutoff: June 8, 2026
  - EASI efficacy (placebo-adjusted % change; observed between-group differences, p-values from MMRM): wk1 –35.56% (p wk2 –61.10% (p<0.0001); wk4 –63.46% (p<0.0001); wk6 –78.95% (p<0.0001). EASI reduction continued to deepen throu remained sustained for patients followed
  - Additional exploratory endpoints (placebo-adjusted, observed between-group differences): EASI-50 — wk1 25%, wk2 75% wk6 82%. EASI-75 — wk1 0%, wk2 17%, wk4 45%, wk6 64%. PP-NRS % change — wk1 –29%, wk2 –35%, wk4 –41%, wk NRS improvement began as early as day 1 and was sustained for 8 weeks after the last dose
- Biomarkers: Early and substantial reductions in TARC and IgE, sustained 8 weeks post-last dose — confirming durable Type 2 pa suppression
- Safety: BBT001 was well tolerated. No conjunctivitis reported (consistent with prior healthy volunteer findings). Low immunogen ADA titers, no apparent neutralizing activity
- PK: Extended half-life in AtD patients consistent with healthy volunteer data, supporting potential maintenance dosing as infrequ every 3 months (Q3M)
- Key upcoming catalysts: Full results at upcoming medical conference (TBD). 1H 2027 — topline 12-week IV data in bio-naïve and experienced AtD; 14-week IV data in CSU. Phase 2b initiation in moderate-to-severe AtD evaluating SC maintenance dosing up t
- J&J terminated its IL-4R/IL-31 after an interim analysis of a phase 2b with dupi as comparator. The company said the molecule "c the high-bar efficacy we established for advancing our clinical development programs for atopic dermatitis." (link to VITAL post)
- Read the data in more detail (link). Detailed cross-trial indirect comparison data are presented in the chart below

Details

- Trial design: This study is part of a global, randomized, double-blind, placebo-controlled Phase 1 clinical trial (NCT06808477) ev IV and SC formulations of BBT001 in healthy volunteers and AtD patients. The AtD portion enrolled bio-naïve patients with mode severe disease (no prior biologics targeting the same pathways as BBT001, no prior JAKi), randomized 2:1 to BBT001 450mg IV placebo during a 4-week treatment period at sites in New Zealand and the United States.
- Baseline Characteristics:

Parameter	BBT001 (n=12)	Placebo (n=

EASI, mean (SD)	34.64 (11.38)	29.06 (14.0
BSA %, mean (SD)	62.58 (19.87)	47.60 (25.1
Weekly PP-NRS, mean (SD)	7.34 (0.87)	6.33 (1.55
vIGA, mean (SD)	3.42 (0.51)	3.20 (0.44

- Efficacy — EASI reduction (placebo-adjusted): Analysis: Observed between-group differences; p-values from MMRM

Visit	Placebo-Adj. Change (%)	95% CI	
Week 1	−35.56	(−50.96, −13.59)	
Week 2	−61.10	(−81.24, −43.87)	
Week 4	−63.46	(−80.75, −43.27)	
Week 6	−78.95	(−94.62, −57.14)	

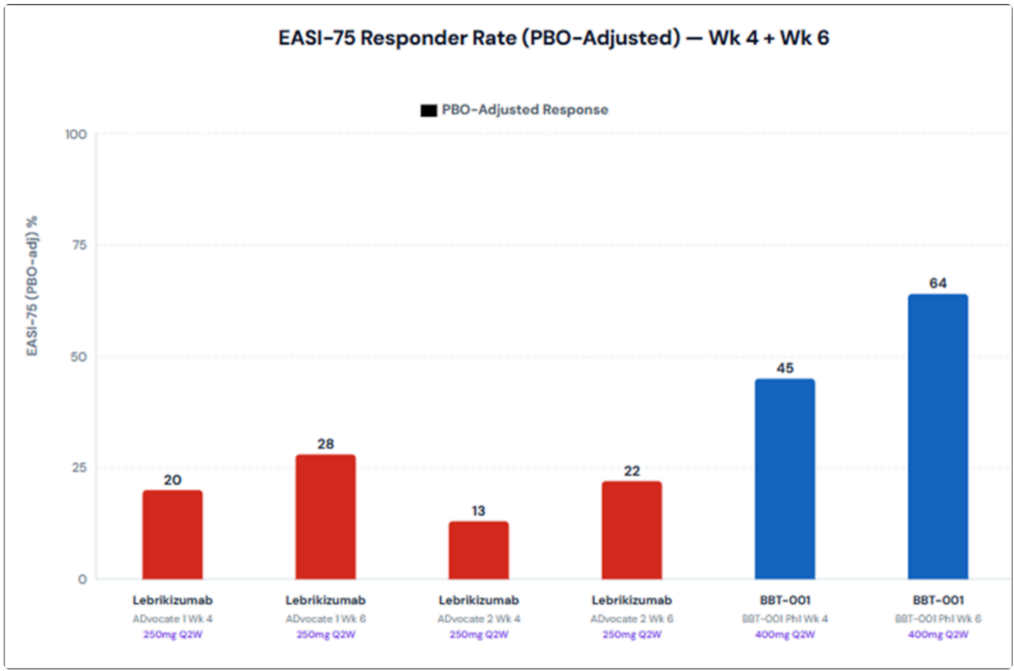
- Exploratory endpoints (placebo-adjusted)

Visit	EASI-50 (%)	EASI-75 (%)	PP-NRS Change (%)
Week 1	25	0	−29
Week 2	75	17	−35
Week 4	91	45	−41
Week 6	82	64	−43

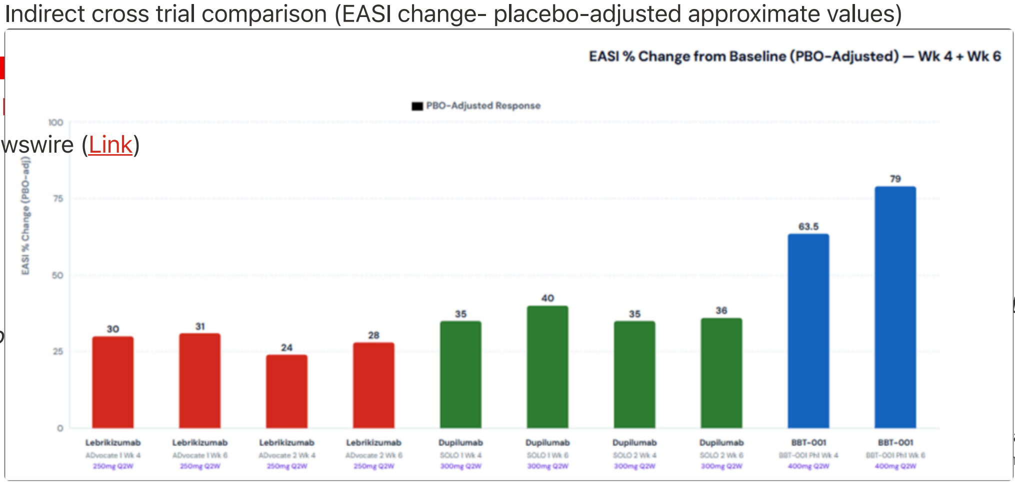
- Biomarkers: BBT001 produced early and substantial reductions in TARC and IgE that were sustained for 8 weeks after the last dc demonstrating durable inhibition of Type 2 inflammation.
- Safety: BBT001 was well tolerated. No cases of conjunctivitis were reported, consistent with prior findings in healthy volunteers.
- Immunogenicity: Low incidence of treatment-emergent anti-drug antibodies, with low titers and no apparent evidence of neutral consistent with prior healthy volunteer findings.
- PK: BBT001 demonstrated an extended half-life in AtD patients, consistent with prior observations in healthy volunteers, suppor potential for maintenance dosing as infrequently as once every 3 months

Observations/Comments:

- BBT001 is a novel tetravalent “2+2” bispecific antibody engineered to simultaneously target the IL-4/IL-13 and IL-31 pathways, fe modifications to extend its half-life
- Company pipeline: BBT002 (IL-4Rα/IL-5 — asthma, COPD, CRSwNP; Phase 1b/2a); BBT003 (GI, preclinical); BBT004 (autoimmu preclinical)
- Data caveats — interpret with caution:
  - Very small N (17): Only 5 placebo patients
  - Baseline severity imbalance: Active arm entered with higher disease burden (EASI 34.64 vs 29.06; BSA 62.58% vs 47.60% to the mean could inflate observed treatment effects. The 95% CIs are wide accordingly
  - IV formulation only: 450 mg IV Q2W is not a commercial-stage dosing regimen. The Q3M maintenance dosing claim is bas extrapolation, not clinical demonstration at that dosing interval. SC PK/efficacy — critical for commercial viability — remain
  - Bio-naïve only: No bio-experienced patients in this cohort. Real-world AtD populations increasingly include patients with p exposure. Bio-experienced IV data expected 1H 2027
  - Exploratory endpoints: All efficacy endpoints are exploratory; the primary endpoint is safety/tolerability. Not powered for ef conclusions.
- Indirect cross trial comparison (EASI-75- placebo-adjusted approximate values)



- Indirect cross trial comparison (EASI change- placebo-adjusted approximate values)



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